MOHAMED BIN ZAYED UNIVERSITY OF ARTIFICIAL INTELLIGENCE

METAVERSE LAB

CV 802: ADVANCED 3D COMPUTER VISION

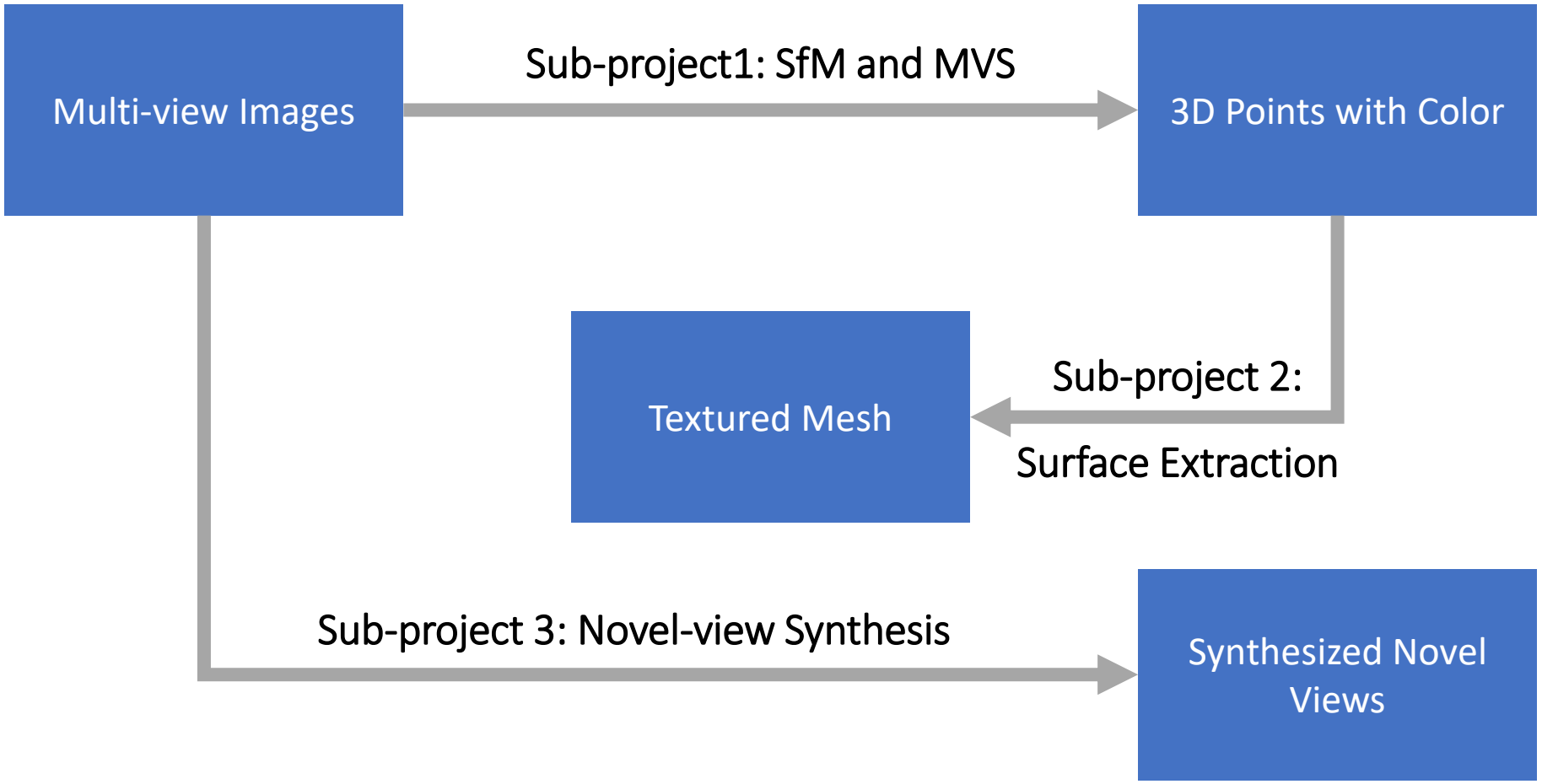
BYO 3D Capture Project Introduction
&
Sub-project 2: Surface Extraction

Paul Zheng

09/18/2025

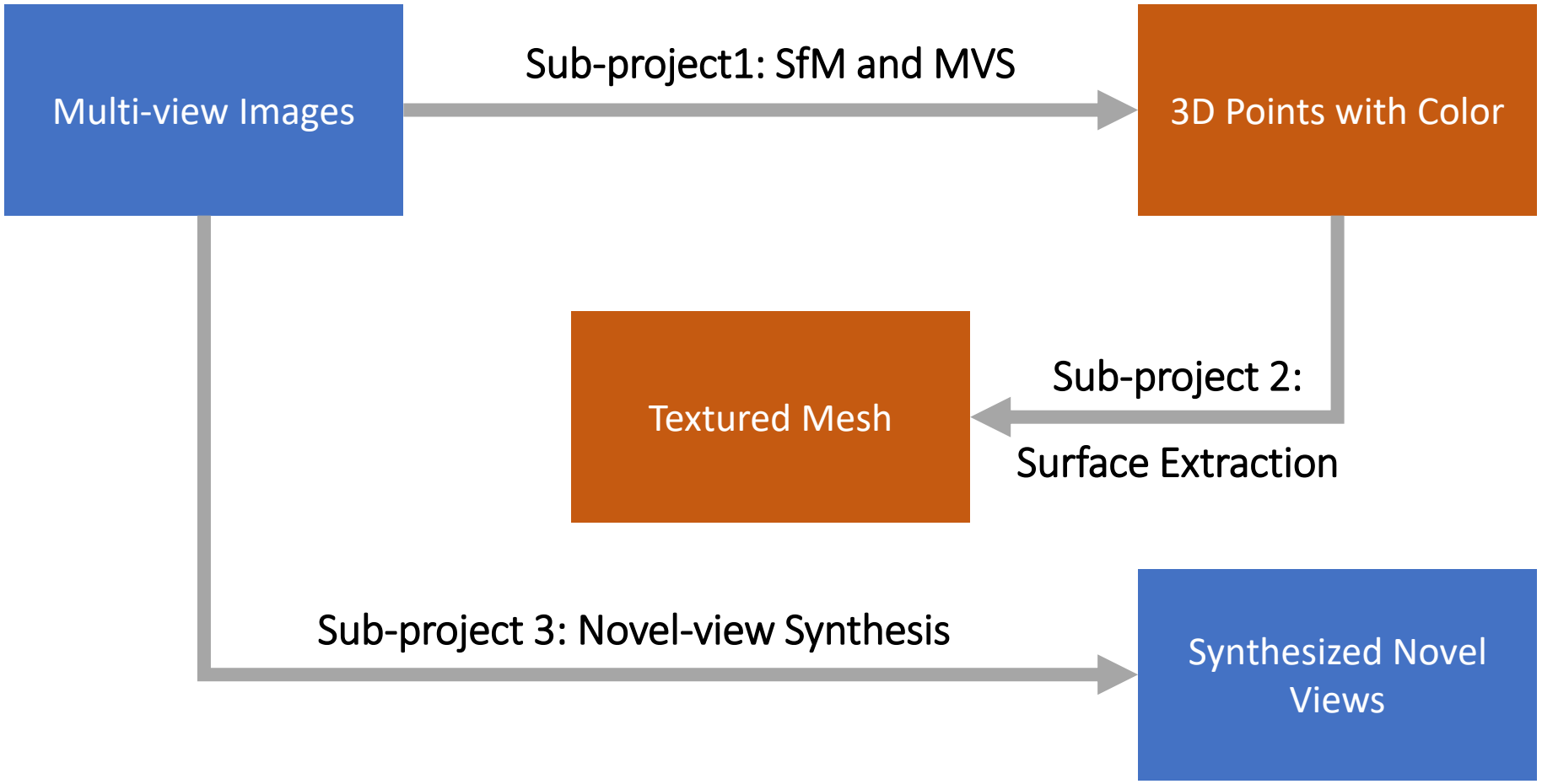
Project Overview

3D Capture System



Project Overview

3D Capture System



Sub-project 2

Surface Extraction

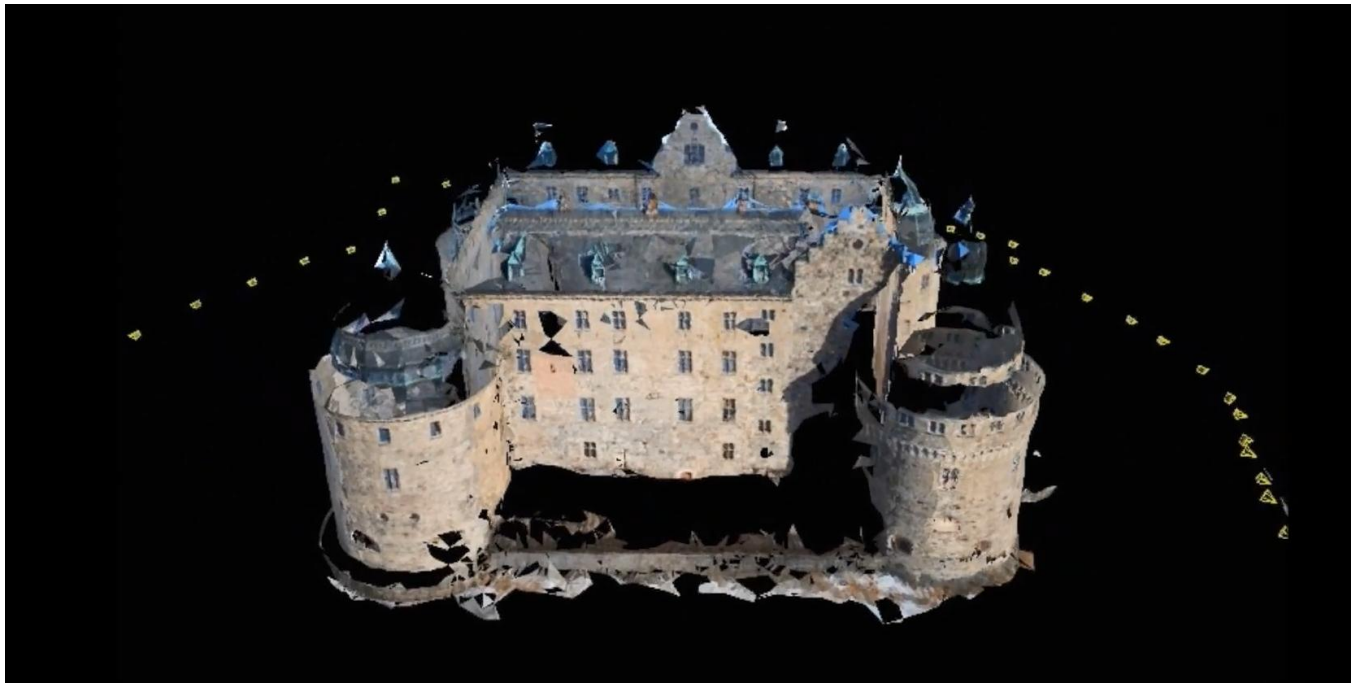
Input



3D Points from Sub-project1

https://www.youtube.com/watch?v=i7ierVkXYa8&t=9s

Output



Textured 3D Mesh

Output format should be .obj
You will get bonus if you provide a Mesh Viewer

https://www.youtube.com/watch?v=i7ierVkXYa8&t=9s

Surface Extraction – Example Methods

Poisson Surface Reconstruction (DO NOT USE API)

- Famous approach
- Easy to be implemented

Meshing Point Clouds with Predicted Intrinsic-Extrinsic Ratio (ECCV2020)

- Good to learn how to combine geometric prior with deep learning

Point2Mesh: A Self-Prior for Deformable Meshes

- Self-supervised, no need to train on large-scale dataset

MeshAnything: Artist-Created Mesh Generation with Autoregressive Transformers

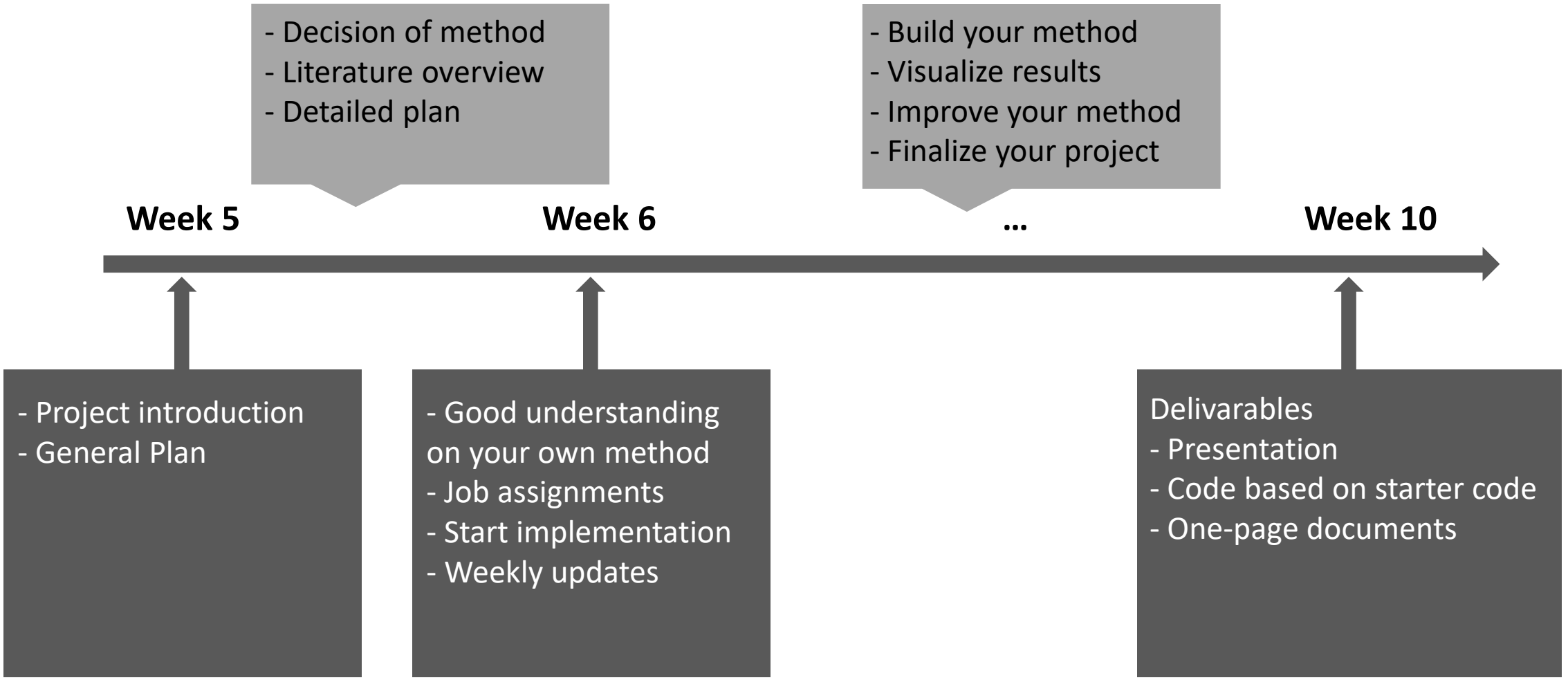
- Recent method with high-quality output, good to have a try
- Only shape (above all methods), you need to get texture by yourself

Deep Hybrid Self-Prior for Full 3D Mesh Generation

- Self-supervised (shape and color), no need to train on large-scale dataset

Your own novel methods!

Timeline of Sub-project 2



For Next Week

Presentation of your chosen method (< 5 minutes)

- Proof you have good understanding on your own method

- Job assignments

- Start weekly updates

Questions?



https://mbzuai.ac.ae/news/applications-now-open-for-new-six-day-intensive-artificial-intelligence-program-for-local-and-international-executives-at-mbzuai/